

Unitree Week 26-28

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Week 26

This Week

- Working on the Observation-Based Reinforcement Learning with the Unitree R1.
- Lesson Plan for Embodied AI club/lab

Set of Lessons/Labs

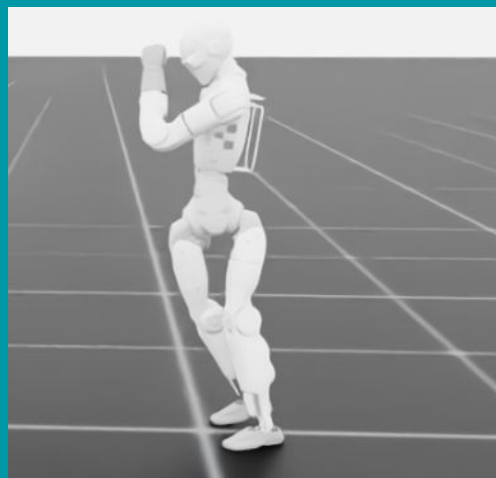
- Reinforcement Learning:
 - IsaacLab + IsaacSim Setup
 - TestRL with basic Unitree G1 locomotion runs (rough_env_cfg / flat_env_cfg)-
 - Unitree G1 training on other terrains (pyramid) → Teaches the need for different penalty uses.
 - Unitree R1 usd file implementation → Teach how to train using outside robot designs
 - Unitree R1 Staircase Training → Teaches to make your own environment setup, your own rewards/penalties.
- Inverse Kinematics:
 - Showcase AMASS → IK
- Extra: If AMASS observation-based learning works, show them this too.

Resolving Issues with Observation-Based-RL

- Robot physically did not always start in the ground in the 0th frame.
 - Integrated offset so that the R1 spawned physically on the ground.
- Implemented imitation-based training.
 - The R1 now is rewarded for following the specific motion (provided by the .npz file) as much as physically possible.
- Speed of motion curriculum:
 - Lowering the speed of motion initially allows for the R1 to have more time to learn the specific motions. As it improves, we increase speed of motion so the end point is mastering the motion at the same speed as the .npz file.

Current Training Results

- The training is now becoming too imitation heavy → focuses primarily on joint positioning while ignoring the physical motion.
- More tuning is required.
- However, should probably test with more training first (ex. 1024 envs with 2000 iterations)



Week 27

More Research on reference-motion RL

- The main issue with the RL pipeline right now is the inability for the robot to pass through the first frame (frame 0). Solutions to this can be to implement:
 - Frame-0 physical feasibility confirmation
 - Foot-contact rewards
 - Future Reference Information
 - Reference State Initialization
 - Low-level actuator sanity

DeepMimic

- How does DeepMimic do RL pipelines:
 - Physical Robot Model that can stand before imitation
 - Reference Motion Library
 - PD-controlled action
 - Joint Tracking
 - Body/Key-point Tracking
 - Early Termination: Timeout, motion complete, non-contact body ground contact, excessive pose deviation.
 - Reference State Initialization (can start an episode from different states sampled from the reference motion).
 - Future Target Observations (Lookahead): Information on the current frame and upcoming frame.

Current Next Steps

- Developing RL standing pipeline (no .npz files involved yet)
 - *Note: this could also be a simple setup to use for the *Embodied AI* course as its simply training the robot to stand.
- Once this works, we work on RL for frame 0 (Going from standing position to frame 0).
- Next, further RL training: training could probably be done in smaller chunks (5-10 .npz file frames for training for example).
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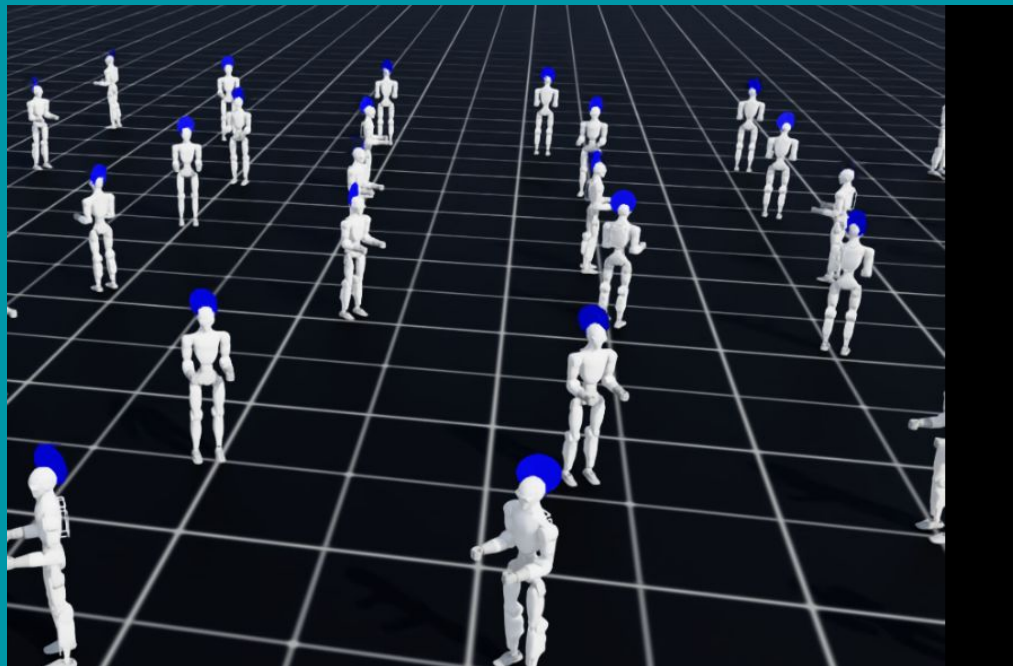
Reusing RL code from rough_env_cfg.py

- R1_EDU_CFG robot asset setup
- Joint Deviation Penalties
- Flat Orientation Reward
- Action-Rate Penalty
- Torque, Acceleration Penalties
- Reset Settings

Developing a model that only teaches the R1 to stand could easily be done using the rough_env_cfg.py code.

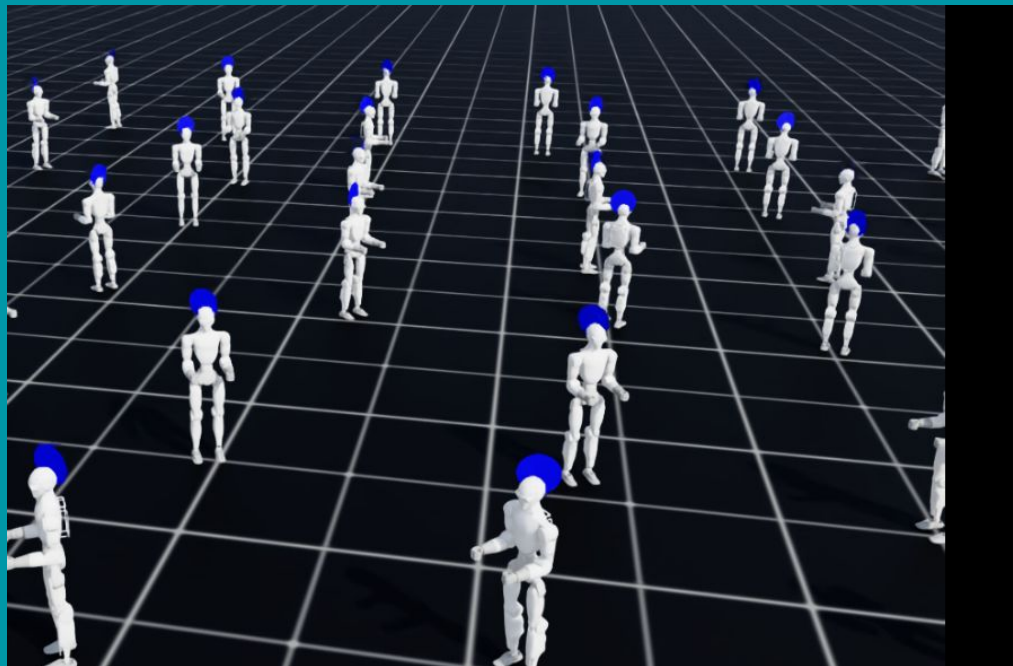
Reusing RL code from rough_env_cfg.py

- After 1000 iterations, the robots appear to perfectly master standing position.
- Now we will try to integrate this initial motion to the motion-reference RL.



Starting Reference-Motion RL with stand_env_cfg model

- Now that we have a RL model with a standing-capable R1, we can use this to then proceed with the first frame of an npz file.
- Standing R1 model $\rightarrow$ Frame 0 $\rightarrow$ Frame 1 $\rightarrow$...



Reference Motion with Stand RL input

- Stage 1: Standing Policy
 - Inputs: 90 observation values
 - Outputs: 26 joint actions
- Stage 2: Reference-motion Policy (requires more information, hence more inputs)
 - Inputs: 202 observation values (current robot pose, reference pose, distance of pose from reference, frame # → hence more than double the inputs of Stage 1)
 - Outputs: 26 joint actions

Reference Motion with Stand RL input

- Since number of inputs of stage 1 is not the same as stage 2, we make the inputs of stage 1 match that of stage 2.
 - Then, using layers, we develop the motion actor.
 - $90 \rightarrow 256 \rightarrow 128 \rightarrow 128 \rightarrow 26$
 - $202 \rightarrow 256 \rightarrow 128 \rightarrow 128 \rightarrow 26$
 - We then append the standing actor onto the motion:
 - Motion first layer [0:90] = standing stage inputs
 - Motion first layer [91:202] = zeros
 - Later actor layers $\rightarrow$ copied from standing policy
 - Actor output layer $\rightarrow$ copied from standing policy
- **The motion policy ignores the .npz inputs initially, allowing it to behave like the standing policy. Once motion stage starts, reference motion inputs are learned in the policy, maintaining the standing's balance behavior.**

Relative Motion Target

- Previously, the .npz was used directly for the exact position from the video.
- Now, the .npz motion is used as a **relative motion target**, meaning the robot learns to do the motion relative to the standing position.
- We also do not have the robot perform the full motion from the start anymore. We start with small motion amplitude, speed, and frame length.
 - Ex.
 - Initially: 10% motion, first 5 frames with slow speed.
 - Next: 25% motion, first 10 frames
 - Then: 50% motion, first 20 frames.
 - Then: 100% motion, all frames
 - Finally: entire clip.

Results

- After training for 500 iterations, the robot failed to learn 0th frame due to failing to adapt to the new training environment.
 - More focus on translating the standing model to the motion-reference model needs to be implemented.
 - Will create an environment that solely focuses on training the R1 from standing environment to the motion-reference environment.
 - `stand_env_90` → `stand_env_202` → `reference_motion_env`
 - This should ensure the robot stands in the 202-input policy environment.

Week 28

Revamp

- The “stand_env90 → stand_env_202” translation was a success.
- For the time being, we will maintain the training code in the IsaacLab R1 directory (the RL code will work directly with the isaacsim RL algorithms; particularly the `rsl_ppo_cfg.py` code.)
 - This should decrease possible variable differences when training with reference-motion RL.



- Visual test of R1 after 1 iteration of `motion_env_cfg.py` (training environment) → signifies it comprehends the `stand_env_cfg`'s learned results.

motion_ref_env_cfg.py

- This file is the training environment file that is used in the IsaacLab R1 directory.
 - First, tested if the training maintains the standing capabilities.
 - Next, test if we can slowly train reference motion with the initial standing capabilities.
- Current problems:
 - Although the robot maintains the standing position with 202 inputs, the standing policy still overpowers the reference-motion policy.
 - It's possible that the standing policy is locked → more research needs to be done.
 - Some other problems that could affect the code that will be tweaked:
 - The number of steps may be too small, causing the robot to prioritize standing much more than the small part of the run wanting to move the joints.
 - The weights of the standing environment far exceeds that of the reference-motion.